A novel multi-resolution method using wavelet packet transform and k-means clustering for analyzing data of passive cavitation detectors

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Acoustic emissions from cavitating bubbles contain complex information in the time and frequency domains. Passive cavitation detectors are used to measure these emissions. By processing this data, it becomes possible to extract cavitation features including broadband emissions, harmonics, sub- and ultra-harmonics. This paper presents a novel approach that utilises the wavelet packet transform, statistical metrics and k-means clustering to detect and classify these features in acoustic emissions. These findings enable us to quantitatively separate the stable and transient features of acoustic cavitation. This method was validated on acoustic emissions obtained by solving a single bubble dynamics (Gilmore-Akulichev model) and subsequently tested on experimental data from biomedical ultrasound applications.

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I. INTRODUCTION

Cavitation refers to the formation and growth of bubbles (cavities filled with gas and vapour) inside a liquid[1]. Acoustic cavitation occurs when the bubbles are created and/or excited using acoustic waves. The bubble nuclei in the liquid grow during rarefaction (i.e. low pressure) cycles of acoustic waves and shrink during compression (i.e. high pressure) cycles. These bubbles may undergo either transient oscillations (known as inertial or transient cavitation) or stable oscillations (known as non-inertial or stable cavitation). In stable cavitation, the bubble continues to oscillate in size until the source is removed. In transient cavitation, however, the bubble grow and implode violently within a few cycles of excitation, resulting sharp localised increases in temperature and pressure. The collapsed bubble can fragment into smaller bubbles which will serve as cavitation nuclei for the subsequent acoustic waves[2]. There are several physical effects associated with these two types of cavitations, including creation of free radicals, shock waves, shear stresses, microjets and micromixing in the liquid, etc.[1]. To utilise these effects for different industrial applications, ultrasound waves are widely used to produce acoustic cavitation, e.g. imaging ultrasound REF, therapeutic ultrasound REF, sono-chemistry, and sonocrystallisation REF. To monitor these processes and to perform mechanistic studies, the acoustic cavitation must be measured and evaluated quantitatively.

Passive REF and active REF cavitation detections are the two main techniques used to quantitatively measure acoustic emissions generated by bubbles. Passive techniques use a single element ultrasound transducer, i.e. passive cavitation detector (PCD), to acquire acoustic emissions whereas active techniques use more sophisticated ultrasound array transducers. Passive methods are cost-effective and easy to implement, hence have been widely used in many different applications REFS.

Acoustic cavitation primarily finds use in biomedical applications. This is because bubbles can affect fluids and tissue inside the body in a cheap and non-invasive manner. Examples include drug delivery and histotripsy. Acoustic cavitation also finds use outside of medicine in applications such as food processing and sonochemistry.

A. Types of Acoustic Cavitation

Cavitation occurs when the static pressure of the liquid falls below its vapour pressure, causing the formation of cavities filled with vapour[1].

In general, ultrasound with low frequencies and high intensity will cause collapse cavitation. Other factors affect this, such as the size and content of the bubble.

B. Project Motivation & Aims

This project aims to implement a method for processing experimental data gathered over a wide range of frequencies of ultrasound (300kHz – 6MHz). This range of frequencies encompasses the types of cavitation that commonly occur. This is important because the nature of the cavitation must be precisely manipulated in many of its applications[2]. The behaviour will also change depending on several conditions e.g., temperature of liquid, gas content of liquid, intensity of ultrasound, etc. but the response of the cavitation to variation in ultrasonic

frequency is especially important to analyse as one of the more easily controlled factors.

II. LITERATURE REVIEW

The methodology behind this project is based on previous works carried out in related fields. Firstly, the signal obtained from PCD is

A. Measurement

B. Features to Extract

The signal from PCD is expected to contain a range of harmonics[1]. These have frequencies which are multiples of the fundamental frequency (the frequency of the ultrasound). Neppiras outlined the expected acoustic emissions for ultrasound at a range of power levels[3]. At low powers, the fundamental frequency was the only one measured. As the power was increased, but before collapse cavitation occurred, high-order harmonics were emitted by the bubbles. The emission with frequency equal to twice fundamental was prominent and had an amplitude proportional to the square of the fundamental.

Subharmonics occasionally appear. These have frequency of 1/N of the fundamental where N is an integer. This can be formed by both stable and inertial cavitation and there are many theories as to what may cause subharmonics[1]. The main theory is that they are generated by the non-linear pulsations of stable cavitation. If they are emitted by a system with only inertial cavitation, this can be explained by the prolonged expansion phase that can occur when the bubble oscillates for a few cycles before collapsing. The 1/2 and 1/3 frequency sub-harmonics are the most commonly occurring ones in acoustic cavitation.

The final component of the signal is broadband noise. There are several theories as to why this might occur. Most importantly, it is an indicator of transient cavitation. The shock wave is thought to cause this[4]. Another possible cause for broadband noise is the bubble-to-bubble interaction, where bubbles pulsate non-periodically[5]. The interaction between bubbles is thought to produce harmonics and sub-harmonics as well, further increasing the difficulty in analysing clusters of bubbles. Broadband noise has the effect of increasing the intensity of each of the harmonics and sub-harmonics. The transducer will also produce some noise as well. The frequency ranges affected depends on the specifications of the transducer. Overall, the feature extraction method must be able to separate and quantify harmonics, subharmonics, and broadband noise. This will make it possible to extract and classify features.

C. Feature Selection

Once data is acquired using PCD, the resulting sound wave is processed using the wavelet transform (see section **??**) to extract features. From the transform, features must be selected. Matloobi and Riahi trained a neural network to identify cavitation inside a centrifugal pump using the discrete wavelet transform to extract features.[6] They used statistical features such as room mean square, kurtosis, skew, mean, and peak of the signal and the transformed data. These features are commonly used among approaches which used k-means[7][8]. These features are calculated in the

D. K-means

After extracting features, the features can be classified using methods such as thresholding and clustering. K-means is a simple algorithm used for clustering. It groups the data into 'k' clusters. The aim of the algorithm is to minimise the total loss measured as the sum of the Euclidean distances between the centroid of the cluster and each point assigned to it (see equation 1. To do this, the cluster centroids are initiated randomly and each data point is assigned to the nearest cluster. Next, the clusters are re-allocated to minimise the loss function. This process of shifting and reassigning the clusters are repeated until convergence.

$$\Theta = \sum_{i=1}^{n} \sum_{j=1}^{k} \rho^{(i,j)} \|x^{(i)} - \mu^{(j)}\|^2 \tag{1}$$

Where Theta is the distortion (loss) function, ρ is the assignment of the data point, i, to cluster, j, and x and μ are the data point and cluster centroid respectively.

K-means is a non-convex optimisation problem and yields different results depending on the starting position of the clusters. There are several methods that have been developed to combat this, explained by Yiakopulos et al.[7].

K-means has been combined with the wavelet transform in applications such as fault detection and EEG prediction. Both of these domains aim to categorise complex signals containing a mixture transients and harmonics. Yiakopulos et al.[7] applied the k-means algorithm to the problem of fault detection for rolling element bearings. Their method used two instances of K-means, one to determine whether the bearing was faulty and another to determine the nature of the fault. Each algorithm used a different set of used a different of features. They noticed that the non-convexity of k-means was a major disadvantage of the k-means algorithm and identified several techniques that have been developed to reduce the effect of this. In their method, the centres were initialised using results from a dynamic model of the system. Shete et al.[8] also used the k-means algorithm to detect the k-complex in EEG signals. This was achieved by using the k-means algorithm to compare feature of a reference k-complex signal and extracted sections of an input signal to determine whether that section contained a k-complex. This allows the training of the algorithm by providing it with examples of the feature that is sought after.

E. Fuzzy C-means

Fuzzy c-means is an extension of k-means where each data point is given a soft cluster assignment $0 < \rho^{(i,j)} < 1$ which represents the degree of membership that a given data point has to each cluster[9]. The advantage of this over k-means is that it allows for the classification of components which overlap with other clusters.

III. METHODOLOGY

This section will outline the methods used by Teh[10] and Haqshenas[11].

A. Wavelet Packets

The short-term fourier transform (STFT) is commonly used in the field of analysing cavitation[10]. This method splits the signal up using windows and computes the Fourier transform on each to add a notion of time to the transform. The equation is given by equation 2. Each set of coefficients will have a fixed time and frequency resolution depending on the selected window size. For large windows, the frequency resolution is high while the time resolution is low. This is the other way around for short windows.

$$X(\tau, \omega) = \int_{-\infty}^{+\infty} x(t) w(t-\tau) e^{-i\omega t} dt \tag{2}$$

Where $w(t)$ is a window function.

The discrete wavelet transform replaces the windowed exponential term with a wavelet (see equation 3). Wavelets are transient, oscillating functions which can be stretched using the scale parameter. This gives the transform a higher frequency resolution for lower frequency components and a higher time resolution for lower frequency components, a much more natural way to break down a signal.

$$F(\tau, s) = \frac{1}{\sqrt{s}} \int_{-\infty}^{+\infty} x(t) \psi \left(\frac{t-\tau}{s} \right) dt \tag{3}$$

Where s is the scale parameter.

The discrete wavelet transform is implemented as a series of low- and high-pass filters, given by h[n] and g[n] respectively (see figure 1). Each set of coefficients is calculated using a convolution given by equation 4. The low-pass filter produces approximation coefficients representing the amplitude of lower frequencies, while the high-pass filter produces detail coefficients, representing higher frequencies. Each set of detail coefficients correspond to a frequency range (subband) of 1 octave. The approximation coefficients correspond to the bottom end of the frequency range.

$$y[n] = (x * g)[n] = \sum_{k=-\infty}^{\infty} x[k] g[n-k] \tag{4}$$

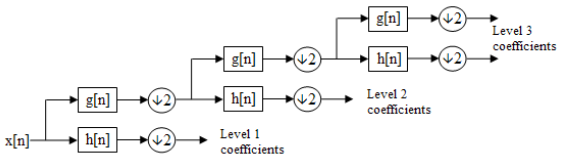


FIG. 1. The DWT implemented as a recursive filter bank

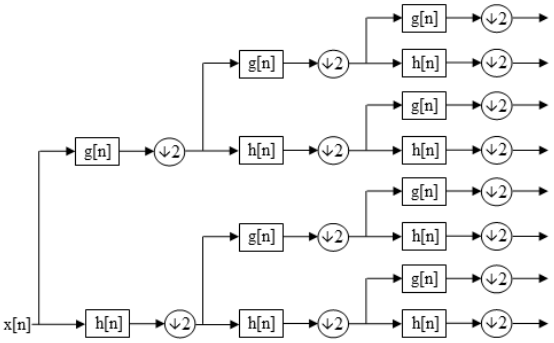


FIG. 2. The DWPT implemented as a filter bank

The discrete wavelet packet transform (DWPT) builds on this by filtering the approximation coefficients as well as the detail coefficients at each level (see figure 2). This has the effect of improving the frequency resolution (and decreasing time resolution) at higher frequencies compared to the DWT. The more levels of transform, the narrower the frequency subbands.

B. Classification of Features

When the wavelet packet transform is applied on a signal, the resulting coefficients at each node correspond to the correlation of the signal with the wavelet. The coefficients at each frequency subband can be separately reconstructed into signals which combine to form the original signal. The stable and transient components of the signal tend to have greatly different frequencies, so the transform will help separate these out. Once separated, the subbands can be classified to give an indication of the presence and extent of each type of cavitation.

1. Statistical Measures

Haqshenas[11] used statistical measures on the wavelet packet coefficients, calculating the energy, kurtosis, and crest factor at each subband. These are able to distinguish between transient and stable signals.

The energy of a subband is a measure of how much the signal is concentrated within it. Energy is calculated as the variance of the coefficients belonging to that subband (see equation 5). This is useful for separating out fundamental frequency and higher amplitude harmonics.

$$E = 1/N v^T v \tag{5}$$

Where v is a vector containing the coefficients in a subband and N is the number of coefficients in the subband.

Kurtosis is a measure of how often outliers occur. This is calculated using the equation 6. High kurtosis is used as an indicator of transient cavitation.

$$K = \frac{E(x-\mu)^4}{\sigma^4} \tag{6}$$

The crest factor is the ratio between the peak value and the mean of the signal (see equation 7. This measures how extreme the peaks are in the signal. A higher crest factor indicates the presence of high amplitude outliers in the signal.

$$C = \frac{\max\left(|x|\right)^2}{x_{rms}^2} \tag{7}$$

These features are calculated on the wavelet coefficients.

2. Analysis

The statistical measures give an indication of the strength and stablity of the signal in each frequency subband, allowing for the classification of stable and transient features. This is combined with thresholding, where the features are selected based on whether their statistical measures are above or below a given value. The illustrate the effect of thresholding, a synthetic signal (see figure 3 was transformed (see figure 4). This data was generated using 2 sine waves with frequencies 1 and 100Hz. A series of short, high amplitude hanning windows were added to this to represent transients. Finally, the signal was windowed. No noise has been added.

The FFT of the signal represents the transients as a range of low amplitude frequencies as shown by the low level of background noise around the two peaks. In the wavelet packet transform, the transients are more visibly separated out across different subbands and more localised time.

Next, the subbands are reconstructed to form waves reconstructed waves with the highest energy, kurtosis, and crest factor are shown in figure 5. The highest energy subband was the lowest (fundamental) frequency. The second highest energy subband (not shown here) was the one containing the other frequency of 10Hz, indicating that energy will tend to pick out the high amplitude low frequency components. The subbands with the highest kurtosis were similar to the one shown in figure 5. They were frequency subbands containing only transient components, indicating that kurtosis is a good measure of transience without any noise present. Finally, the subbands with the highest crest factor picked were those remaining in the lower frequency ranges. These contained some transient components mixed with the lower frequency sine waves.

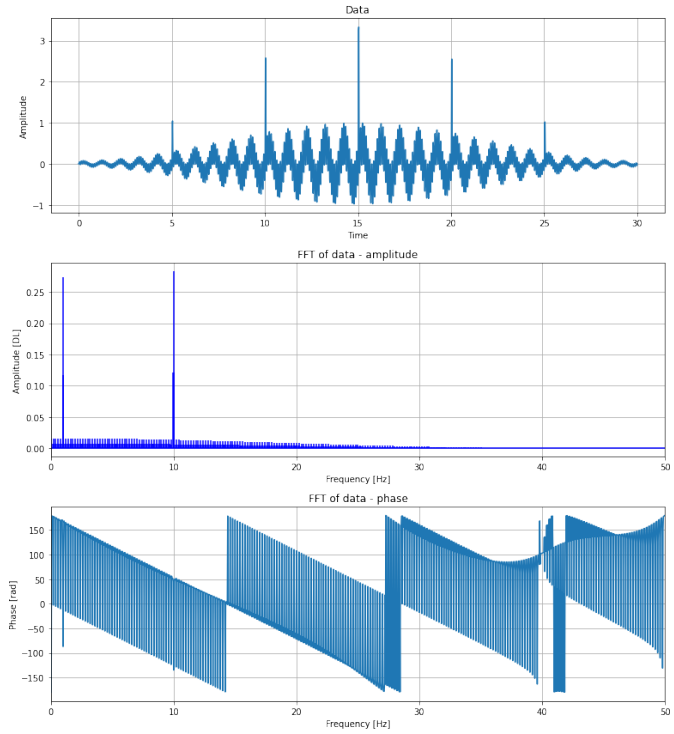


FIG. 3. synthetic data with sampling rate 100 with frequencies 1Hz and 10Hz, transient components, and noise.

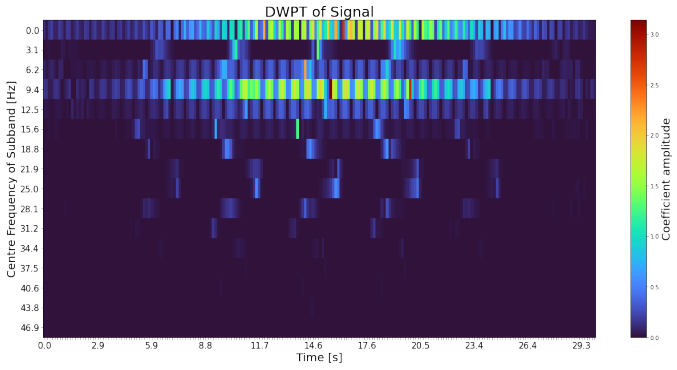
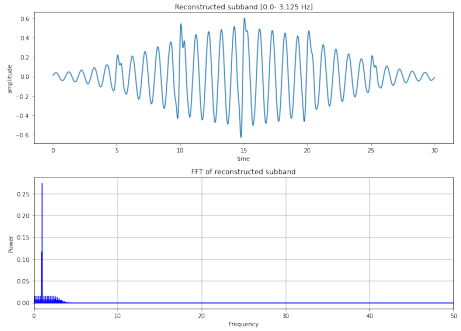
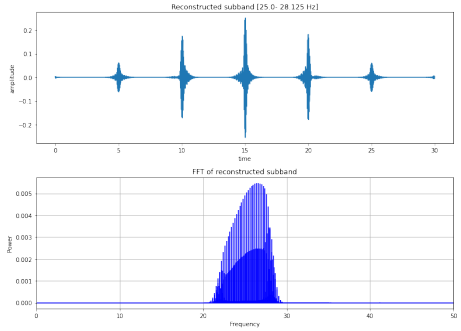


FIG. 4. Heisenberg plot of a 4-level db16 DWPT of a synthetic signal in figure 3.

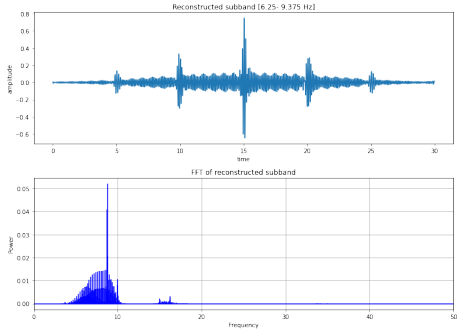
When Gaussian noise is added to the data, the highest subbands sorted by energy and crest factor tends to remain the same, but the measure of kurtosis is impacted and will tend to return similar results to the crest factor.



(a)



(b)



(c)

FIG. 5. Reconstructed subbands with highest energy (a), kurtosis (b), and crest factor (c) respectively

IV. RESULTS

V. SUMMARY

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